

Decentralized Biogas Production with Centralized Upgrading & Grid Injection: When Is It Optimal?

Bachelor of Technology

by

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Abstract

The increasing demand for renewable energy and reduction of greenhouse gas emissions has driven interest in biomass-based energy systems. Among these, biogas production via anaerobic digestion offers a sustainable pathway to produce green gas, which can substitute natural gas.

This study investigates the feasibility of a **decentralized biogas production system with centralized upgrading and grid injection**, based on the model proposed by Hengeveld et al. (2014). An Excel-based computational model was developed to estimate **cost and energy consumption per m³ of green gas** for different system configurations.

Initially, the developed model showed significant deviation from literature values. To resolve this, key parameters such as number of cattle, co-substrate (maize) mass and price, digester investment, upgrading cost, transport distance, and energy consumption were revised. After recalibration, the model results closely matched the reference study.

The validated model was further applied to two real locations in Rajasthan—**Kota and Bikaner**—to analyze geographical impacts on system performance. The results indicate that Kota is more suitable due to lower production cost (62.5 €/t/m³) and lower energy consumption (6.82 MJ/m³), whereas Bikaner shows higher cost (67.9 €/t/m³) and energy requirement (10.33 MJ/m³).

Introduction:

2.1 Background and Motivation

The global energy sector is undergoing a structural transition from fossil-based systems to renewable and sustainable energy sources. This transition is driven by multiple factors, including climate change mitigation, energy security concerns, and the depletion of conventional fossil fuel reserves. Among various renewable energy technologies, **biomass-based energy systems** have gained considerable attention due to their ability to convert organic waste into usable energy forms.

Biomass, derived from agricultural residues, animal manure, and organic waste, represents a **carbon-neutral energy source**, as the carbon dioxide released during its utilization is offset by the carbon absorbed during biomass growth. One of the most efficient pathways for biomass utilization is **anaerobic digestion**, a biological process in which microorganisms break down organic matter in the absence of oxygen to produce biogas.

Biogas typically consists of:

- Methane (CH_4): 50–70%
- Carbon dioxide (CO_2): 30–50%
- Trace gases (H_2S , NH_3 , water vapor)

While raw biogas can be used directly for heat and power generation, its **low calorific value and impurities** limit its applications. Therefore, upgrading biogas to **biomethane (green gas)**—by removing CO_2 , H_2S , and moisture—enhances its quality to match natural gas standards. This upgraded gas can then be:

- Injected into natural gas grids
- Used as vehicle fuel
- Stored and transported efficiently

This capability makes green gas a **strategically important renewable fuel** in achieving decarbonization targets.

2.2 Centralized vs Decentralized Biogas Systems

Biogas production systems can broadly be categorized into two configurations:

1. Centralized Systems

In centralized systems, biomass from multiple sources is transported to a **single large-scale digester**.

Advantages:

- Economies of scale (lower capital cost per unit output)
- Higher process efficiency
- Simplified operation and maintenance

Limitations:

- High transportation cost for biomass
 - Increased logistics complexity
 - Environmental concerns due to transport emissions
 - Risk of system failure affecting entire production
-

2. Decentralized Systems

In decentralized systems, multiple **small-scale digesters** are located near biomass sources (e.g., farms).

Advantages:

- Reduced biomass transport
- Better utilization of local resources
- Increased system resilience
- Lower environmental impact

Limitations:

- Higher capital cost per unit
 - Need for coordination between multiple units
 - Requirement of additional infrastructure (e.g., biogas pipelines)
-

2.3 Hybrid Approach: Decentralized Production with Centralized Upgrading

To overcome the limitations of both configurations, a hybrid system has been proposed (as studied in the reference paper):

Decentralized biogas production combined with centralized upgrading and grid injection

Concept Overview

- Biogas is produced at multiple decentralized digesters
- A **biogas grid (pipeline network)** transports raw biogas
- A central facility performs upgrading and injection

Conceptual Flow

Biomass (Manure + Crop Residue)



Decentralized Digesters



Raw Biogas Collection



Biogas Pipeline Network



Central Upgrading Facility



Green Gas Injection

2.5 Engineering Trade-offs

This hybrid system introduces a complex set of trade-offs:

Advantages

- Reduced biomass transport cost
- Centralized upgrading → economies of scale
- Improved flexibility and modularity
- Better scalability

Disadvantages

- Additional cost for pipeline infrastructure
- Energy consumption for gas compression

- Pressure drop and operational challenges in pipelines

According to the reference study :

- Centralized systems generally have **lower production cost**
 - Decentralized systems do **not show significant energy advantage**
-

2.6 Research Gap

Although several studies have analyzed biogas systems, most of them focus on:

- Single digester systems
- Isolated cost or energy analysis
- Region-specific case studies

The key gaps identified are:

- Lack of integrated models combining **cost and energy analysis**
 - Limited validation of decentralized biogas grid systems
 - Insufficient real-world application studies
-

2.7 Objectives of the Present Study

This project aims to address the above gaps through:

Primary Objectives

1. Develop a **computational model (Excel-based)** to estimate:
 - Cost per m³ of green gas
 - Energy consumption per m³
 2. Validate the model using literature data
 3. Improve model accuracy by refining key parameters
-

Secondary Objectives

4. Apply the model to real locations (Kota and Bikaner)
5. Compare performance based on:

- Cost
 - Energy
 - Transport logistics
6. Identify the most suitable location for implementation
-

2.8 Scope of the Study

The study includes:

- Biogas production from manure and co-substrate
- Transport via biogas grid
- Centralized upgrading and injection
- Cost and energy evaluation

Limitations

- Assumes steady-state operation
 - Does not include dynamic demand variation
 - Simplified transport modeling
-

2.9 Significance of the Study

This work provides:

- A validated engineering model for green gas systems
- Insight into decentralized vs centralized trade-offs
- Practical evaluation for Indian conditions (Kota & Bikaner)

It also demonstrates how **process modeling + real data** can be combined to support engineering decision-making.

3. Definitions and Terminology

3.1 Introduction to Terminology

A clear understanding of technical terminology is essential for analyzing and modeling the biogas-to-green gas production system. The present study involves multidisciplinary concepts from **biochemical engineering, thermodynamics, transport phenomena, and economic analysis**.

This section defines and explains all key terms used in:

- The reference study
- The developed Excel model
- Cost and energy calculations

Each term is explained with:

- Definition
 - Engineering significance
 - Role in the present study
-

3.2 Biomass

Definition

Biomass refers to organic material derived from plants and animals that can be used as a fuel source. In this study, biomass includes:

- Animal manure
- Agricultural residues
- Energy crops (e.g., maize silage)

Engineering Significance

Biomass acts as the **primary feedstock** for anaerobic digestion. Its composition determines:

- Biogas yield
- Methane content
- Process efficiency

Role in Model

- Determines **input mass flow rate**
 - Directly affects **biogas production rate**
 - Influences **transport cost and energy**
-

3.3 Manure

Definition

Manure is organic waste from livestock, primarily consisting of:

- Undigested organic matter
- Water
- Microbial biomass

Engineering Significance

- Provides **baseline substrate** for digestion
- Contains lower methane potential compared to co-substrates

Role in Model

- Used as a **continuous feedstock source**
 - Determines digester loading rate
 - Included in transport calculations
-

3.4 Co-substrate

Definition

A co-substrate is an additional organic material added to manure to enhance biogas production. In this study, **maize silage** is used.

Engineering Significance

- Increases **methane yield**
- Improves **C/N ratio** (Carbon to Nitrogen ratio)
- Enhances microbial activity

Role in Model

- Affects **biogas production rate**
 - Contributes to **total cost (purchase + transport)**
 - Major variable adjusted during model calibration
-

3.5 Anaerobic Digestion

Definition

Anaerobic digestion is a biological process where microorganisms break down organic matter in the **absence of oxygen**, producing biogas.

Stages of Digestion

1. Hydrolysis
2. Acidogenesis
3. Acetogenesis
4. Methanogenesis

Engineering Significance

- Core process of the system
- Determines **conversion efficiency**
- Impacts energy output

Role in Model

- Converts biomass → biogas
 - Defines **digester capacity**
 - Affects cost (capital + operation)
-

3.6 Biogas

Definition

Biogas is the gaseous product of anaerobic digestion, mainly composed of:

- Methane (CH₄)

- Carbon dioxide (CO₂)

Engineering Significance

- Raw fuel with lower calorific value
- Requires upgrading for grid use

Role in Model

- Intermediate product
 - Used in:
 - Transport calculations
 - Compression energy calculations
-

3.7 Green Gas (Biomethane)

Definition

Green gas is upgraded biogas with high methane purity (~95%), equivalent to natural gas.

Engineering Significance

- Can be injected into gas grid
- Higher calorific value
- Suitable for storage and transport

Role in Model

- Final output
 - Basis for:
 - Cost per m³
 - Energy per m³
-

3.8 Digester

Definition

A digester is a reactor where anaerobic digestion takes place.

Engineering Significance

- Determines system scale
- Affects capital investment
- Influences operational efficiency

Role in Model

- Number of digesters = key variable
 - Determines decentralized vs centralized configuration
-

3.9 Decentralized Digestion

Definition

Biogas production using multiple small-scale digesters located near biomass sources.

Engineering Significance

- Reduces biomass transport
- Increases flexibility

Role in Model

- Configuration variable (e.g., 8 digesters)
 - Impacts:
 - Transport cost
 - Pipeline length
-

3.10 Centralized Upgrading

Definition

A single facility where biogas from multiple digesters is processed and upgraded.

Engineering Significance

- Achieves economies of scale
- Reduces upgrading cost per unit

Role in Model

- Fixed cost component

- Same for all configurations
-

3.11 Biogas Grid (Pipeline Network)

Definition

A network of pipelines used to transport raw biogas from decentralized digesters to a central upgrading facility.

Engineering Significance

- Enables hybrid system
- Introduces:
 - Pressure drop
 - Compression energy

Role in Model

- Major contributor to:
 - Cost
 - Energy consumption
-

3.12 Pressure Drop

Definition

Reduction in pressure of biogas as it flows through the pipeline due to friction and flow resistance.

Engineering Significance

- Determines compressor requirements
- Affects energy consumption

Role in Model

- Used to calculate:
 - Compression work
 - Energy consumption
-

3.13 Compressor

Definition

A device used to increase the pressure of biogas for transport through pipelines.

Engineering Significance

- Required to overcome pressure drop
- Major energy-consuming unit

Role in Model

- Calculates:
 - Energy consumption
 - Operating cost
-

3.14 Upgrading Process

Definition

Process of removing CO₂, H₂S, and impurities from biogas.

Engineering Significance

- Improves methane concentration
- Makes gas grid-compatible

Role in Model

- Major energy consumption block
 - Significant cost component
-

3.15 Transport Distance

Definition

Distance between:

- Biomass source and digester
- Digester and central facility

Engineering Significance

- Directly affects:
 - Transport cost
 - Energy consumption

Role in Model

- Key parameter modified during calibration
 - Major difference between Kota and Bikaner
-

3.16 Capital Expenditure (CAPEX)

Definition

Initial investment required for:

- Digester
- Pipeline
- Upgrading facility

Engineering Significance

- Determines long-term cost

Role in Model

- Converted into annual cost using depreciation
-

3.17 Operating Expenditure (OPEX)

Definition

Recurring cost including:

- Energy
- Maintenance
- Labor

Engineering Significance

- Determines operational feasibility

Role in Model

- Added to total cost per m³
-

3.18 Specific Energy Consumption

Definition

Energy required per unit volume of green gas produced (MJ/m³).

Engineering Significance

- Key performance indicator

Role in Model

- Used to compare:
 - Configurations
 - Locations
-

3.19 Cost per Unit Volume (€/m³ or €ct/m³)

Definition

Total cost required to produce 1 m³ of green gas.

Engineering Significance

- Primary economic metric

Role in Model

- Final output used for comparison
-

3.20 Pipeline Diameter

Definition

Internal diameter of pipeline used for gas transport.

Engineering Significance

- Affects:
 - Pressure drop
 - Investment cost

Role in Model

- Optimization parameter
-

3.21 Reynolds Number (Re)**Definition**

A dimensionless number used to determine flow regime:

- Laminar or turbulent

Engineering Significance

- Determines friction factor

Role in Model

- Used in pressure drop calculation
-

3.22 Friction Factor (λ)**Definition**

Dimensionless parameter representing resistance to flow in pipe.

Engineering Significance

- Determines pressure drop

Role in Model

- Used in pipeline calculations
-

3.23 Methane Yield**Definition**

Volume of methane produced per unit mass of biomass.

Engineering Significance

- Indicates efficiency of digestion

Role in Model

- Determines gas production

3.24 Digestate

Definition

Residual material left after anaerobic digestion.

Engineering Significance

- Can be used as fertilizer
- Requires transport

Role in Model

- Included in transport cost

3.25 Gas Injection (Grid Injection System)

Definition

Gas injection refers to the process of introducing upgraded biogas (biomethane/green gas) into an existing **natural gas distribution or transmission network**.

Engineering Significance

Injection is a **critical final stage** in the biogas-to-green gas value chain because:

- It enables **large-scale distribution** using existing infrastructure
- Eliminates the need for dedicated storage or transport systems
- Allows integration with national energy systems

However, injection requires strict compliance with **gas quality standards**, including:

- Methane purity (>95%)
- Controlled moisture content
- Absence of corrosive gases (e.g., H₂S)
- Specific calorific value

Key Components of Injection System

1. **Final Compression Unit**

- Raises gas pressure to match grid pressure
- Typical grid pressure: medium to high (depending on network)

2. **Gas Quality Monitoring System**

- Ensures compliance with grid standards

3. **Metering System**

- Measures volume of injected gas

4. **Safety Systems**

- Shut-off valves
 - Pressure regulators
 - Emergency venting
-

Role in Your Model

In your Excel model and reference study :

- Injection is treated as part of the **“Upgrading & Injection” block**
- Cost includes:
 - Compression energy
 - Equipment cost
- Energy consumption includes:
 - Final gas compression

Important Insight

The research paper assumes:

Injection energy is relatively constant across configurations

Hence:

- It does **not significantly influence comparison between systems**
 - But contributes to **total energy consumption (~fixed component)**
-

3.26 Biomethane vs Green Gas

Biomethane – Technical Definition

Biomethane is:

A purified form of biogas consisting primarily of methane (CH₄), typically >95% purity, obtained after removing CO₂, H₂S, and moisture.

Green Gas – System-Level Definition

Green gas refers to:

Biomethane that is suitable for and actually integrated into the natural gas grid or used as a renewable substitute for fossil natural gas.

Key Difference (Very Important for Thesis)

Aspect	Biomethane	Green Gas
Nature	Chemical composition	Functional/market term
Definition	Upgraded biogas	Grid-quality renewable gas
Focus	Purity & composition	Usability & application
Stage	After upgrading	After injection/use
Scope	Process-level	System-level

Engineering Interpretation

- **Biomethane** → Product of upgrading process
- **Green gas** → Usable energy carrier integrated into energy system

Biogas → Upgrading → Biomethane → Injection → Green Gas

↓

Digestate (by-product)

4. Methodology

4.1 Introduction

The entire model is implemented in the **Biogas Grid Excel sheet**, where each sheet represents a process block. For each sheet:

1. **All parameter values used are explicitly listed**
2. **Step-by-step calculations are performed using these values**

The final outputs are:

- Cost (€/m³ or €/t/m³)
- Energy consumption (MJ/m³)

4.2 Input Parameters

System Configuration

Number of Digesters	8
Biogas per digester	254.5 m ³ /year
Operating hours	8000 hr/year
Total Green Gas	9,600,000 m ³ /year

Feedstock

Number of Cows	4064
Manure Production	20 ton/cow/year
Co-substrate	205,000 ton/year
Co-substrate price	13.5 €/ton

Transport

- Distance = **10.6 km**
 - Truck cost = **1.24 €/km**
 - Fuel energy = **1.3 MJ/(ton·km)**
-

Grid

- Radius = **9.98 km**
 - Pipeline diameter = **110 mm**
 - Cost = **88 €/m**
-

Injection

- Distance = **1450 m**
 - Cost = **130 €/m**
-

4.3 Sheet 2: Biomass Calculation

Manure Production

$$\text{Manure} = \text{No. of cow} \times \text{Manure per cow} = 4064 \times 20 = 81,280 \text{ ton/year}$$

Manure Production	81,280 ton/year
Co-substrate Production	205,000 ton/year
Total Biomass	286,280 ton/year

Interpretation

This total biomass is fed into the digester.

4.4 Sheet 3: Biogas Production (Digester Biogas Sheet)

Parameter Values

Manure Yield	25 m ³ /ton
Co-substrate Yield	175 m ³ /ton
Methane Fraction	0.556

Calculation

	Production	Biogas Yield		Total Biogas
Manure	81280	25	Production*Yield	2,032,000 m ³ /year
Co-substrate	205000	175	Production*Yield	35,875,000 m ³ /year
		Total Biogas Production		37,907,000 m ³ /year

Methane Production

$$\begin{aligned}
 &= \text{Total Biogas Production} \times \text{Methane Fraction} \\
 &= 37,907,000 \times 0.556 = 21,076,290 \text{ m}^3/\text{year}
 \end{aligned}$$

Interpretation

This methane becomes **green gas after upgrading**.

4.5 Sheet 4: Digester Cost & Energy

Parameter Values

Number of Digesters	8
Investment per Digesters	2,025,000 €
Lifetime	12 years
OPEX	5%

Step-by-Step Calculation

Step 1: Total Investment in 8 Digesters

$$= 2,025,000 \times 8 = 16,200,000 \text{ €}$$

Step 2: Depreciation per year

$$= \frac{16,200,000}{12} = 1,350,000 \text{ €/year}$$

Step 3: OPEX

$$= 0.05 \times 16,200,000 = 810,000 \text{ €/year}$$

Step 4: Total Annual Cost

$$= 1,350,000 + 810,000 = 2,160,000 \text{ €/year}$$

Step 5: Cost per m³

$$= \frac{\text{Total Annual cost}}{\text{Total Green Gas}} = \frac{2,160,000}{9,600,000} = 0.225 \text{ €/m}^3$$

Energy Values

Component	Value
Heating	1.194 MJ/m ³
Mixing	0.03 MJ/m ³
Pumping	0.015 MJ/m ³
Total	1.239 MJ/m ³

4.6 Sheet 5: Transport (Manure + Maize)

Parameter Values

- Distance = **10.6 km**
 - Fuel energy = **1.3 MJ/(ton·km)**
 - Cost = **1.24 €/km**
-

Step-by-Step Calculation

Step 1: Ton-km

$$= \text{Manure Production} \times \text{Distance}$$

$$= 81,280 \times 10.6 = 862,330 \text{ ton-km}$$

Step 2: Energy

$$= \text{Manure Production} \times \text{Distance} \times \text{Fuel Energy}$$

$$= 81,280 \times 10.6 \times 1.3 = 1,121,029 \text{ MJ/year}$$

Step 3: Energy per m³

$$= 0.117 \text{ MJ/m}^3$$

Step 4: Cost

$$= 71,286 \text{ €/year}$$

4.7 Sheet 6: Digestate Transport

Parameter Values

- Total digestate = **286,280 ton/year**
-

Calculation

Energy

$$= 3,948,429 \text{ MJ/year}$$

$$= 0.411 \text{ MJ/m}^3$$

Cost

$$= 251,080\text{€}/\text{year}$$

$$= 2.61\text{€ct}/\text{m}^3$$

4.8 Sheet 7: Biogas Grid (Pipeline + Compression)

Parameter Values

- Radius = **9.98 km**
 - Diameter = **110 mm**
 - Cost = **88 €/m**
 - Total length = **139.72 km**
-

Step-by-Step Calculation

Pipeline Cost

$$= 139,720 \times 88 = 12,295,360\text{€}$$

Maintenance

$$= 2\% = 245,907\text{€}/\text{year}$$

<i>required investments:</i>			
Desulfurization + drying	5000	€	per digester
total	40000	€	=8*5000
Flare cost	100000	€	
Total additional cost:	1,40,000	€	

Compression

Compressor cost per unit	148.64305	€	paper formula=111.257+0.1469*C
OPEX	7.4321525		

$$\text{Total Compressor cost} = (148.64305 + 7.4321525) * 7 \{\text{cost per compressor} * \text{compressor needed}\}$$

$$= 1092.526418 \text{ €}$$

Total annual biogas grid cost (with 12-year depreciation)

$$= 1056863.311 \text{ €/year} \quad (\text{pipeline} + \text{maintenance} + \text{additional cost} + \text{compression cost})$$

$$\text{Cost per m}^3 = 0.064886009 \text{ €/m}^3$$

ENERGY CALCULATION:

Inputs

suction pressure(p1)	0.101325	MPa
discharge pressure(p2)	0.32915	MPa
Pressure drop	0.06325	MPa

Uses thermodynamic equation:

$$W = \frac{ZRT}{M} \cdot \frac{k}{k-1} \left[\left(\frac{P_2}{P_1} \right)^{\frac{k-1}{k}} - 1 \right]$$

$$\text{Energy use per m}^3 = 0.272779541 \text{ MJ/m}^3$$

$$\text{Energy cost per m}^3 = 0.010608093 \text{ €/m}^3$$

So,

$$\text{TOTAL BIOGAS GRID COST} = (0.064886009 + 0.010608093) = 0.075494102 \text{ €/m}^3$$

$$= 7.549410202 \text{ €ct/m}^3$$

4.9 Sheet 8: Upgrading

Parameter Values

- Investment = **2,764,000 €**
 - Energy = **0.408 kWh/m³ = 1.4688 MJ/m³**
-

Calculation

$$E = 1.4688 \text{ MJ/m}^3$$

4.10 Sheet 9: Injection

Parameter Values

- Distance = **1450 m**
 - Cost = **130 €/m**
-

Calculation

Investment

$$= 1450 \times 130 = 188,500 \text{ €}$$

Annual Cost

$$= 38,328 \text{ €/year}$$

Cost per m³

$$= 0.399 \text{ €/m}^3$$

4.11 Final Aggregation Sheet

Total Cost

Sum of:

- Co-substrate
 - Digester
 - Transport
 - Grid
 - Upgrading
 - Injection
-

Total Energy

Sum of:

- Digester
 - Transport
 - Digestate
 - Compression
 - Upgrading
-

4.12 Flowchart

Input Parameters



Biomass Calculation



Biogas Production



Digester Cost & Energy



Transport Calculation



Grid Calculation



Upgrading & Injection



Final Cost & Energy